

Worksheet 4: Monopoly

You may work together in groups to discuss and solve these problems, but each person is required to write and submit their own answers. Use the separate answer page to write answers. Each question is worth 2 points. Partial credit may be given.

Part A

Suppose some unfortunate legislation in the burrito industry enables exactly one firm to operate in the market. The inverse market demand for burritos is given by

$$p = 750 - 2.5Q$$

where Q is the output of the monopolist and p is the market price. The firm has a cost function:

$$C(Q) = 5Q^2 + 12,750.$$

1. Find the firm's profit-maximizing quantity.
2. Find the firm's profit-maximizing price.
3. How much profit does the firm make?
4. What is the socially efficient quantity?
5. Find the Deadweight Loss from monopoly.
6. At the monopoly price and quantity, compute the price elasticity of demand. Is the demand elastic, inelastic, or unit elastic? Recall that $E = (dQ/dp)(p/Q)$ where $dQ/dp = 1/(dp/dQ)$.
7. Compute the Price-Cost Markup (Lerner Index). Recall that $L = (p - MC)/p$.
8. Suppose the government decides to regulate the price of burritos. To do so, they set a price ceiling. What price ceiling would lead to the efficient output in the market?

Part B

A large multinational corporation, PzzaCoTM has a new endeavor of buying all of the pizza restaurants in Prince George's County. They even get the county government to prevent other companies from operating in the county. Using the graph of the pizza demand and costs (provided separately), answer the following questions.

9. What quantity maximizes PzzaCoTM's profit?
10. What price maximizes PzzaCoTM's profit?
11. What is PzzaCoTM's maximum profit?
12. What is the DWL from the pizza monopoly?
13. Frustrated by the corporate greed of PzzaCoTM, PGC government decides to regulate pizza prices in the county. They decide to set a price ceiling at a level that maximizes the consumer surplus while assuring PzzaCoTM a non-negative profit. What price ceiling do they set and how many pizzas are sold?
14. What is the deadweight loss from this price ceiling?